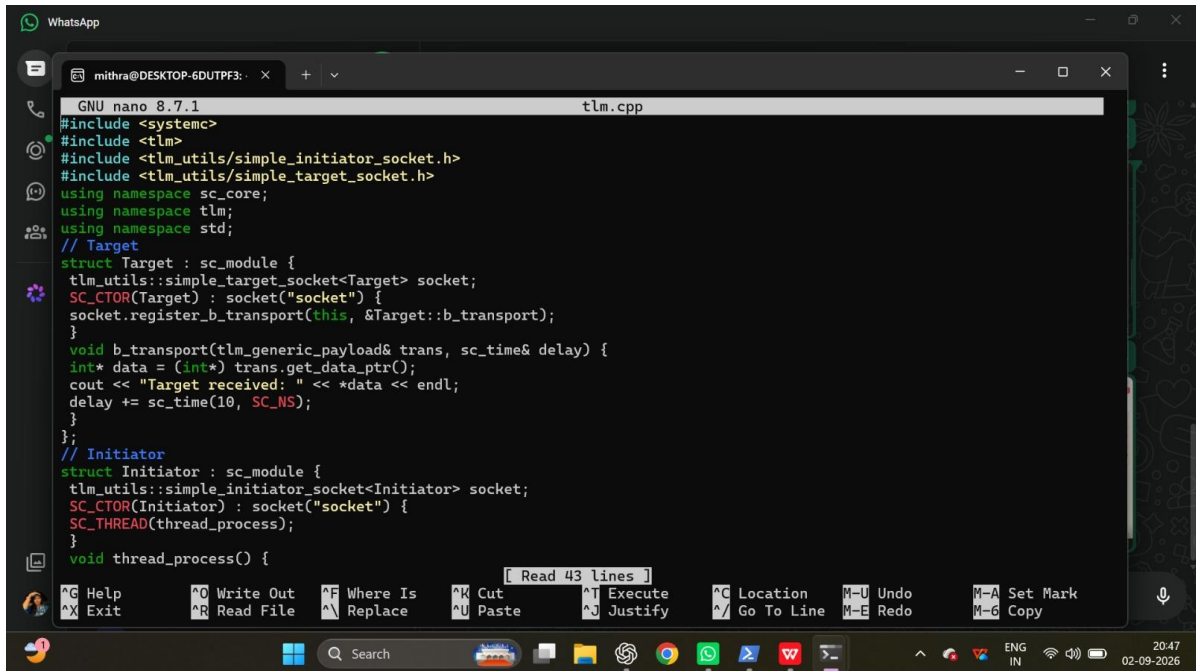
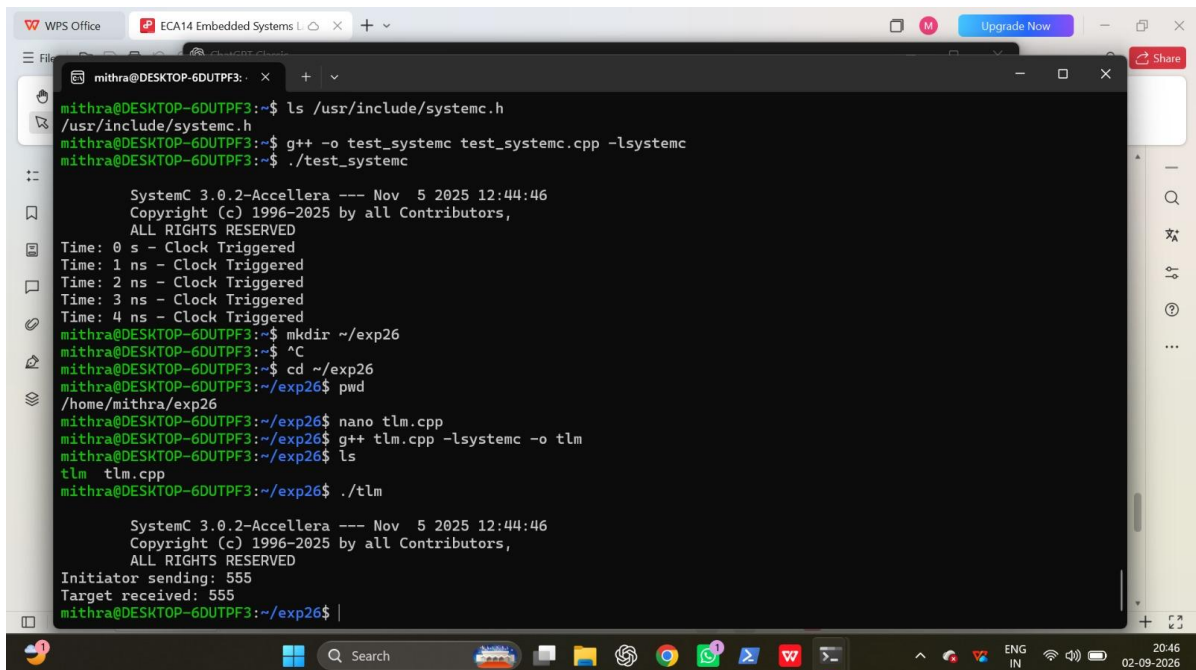


Exp No: 26

TRANSACTION LEVEL MODELING (TLM) USING SYSTEMC



```
GNU nano 8.7.1 tlm.cpp
#include <systemc>
#include <tlm>
#include <tlm_utils/simple_initiator_socket.h>
#include <tlm_utils/simple_target_socket.h>
using namespace sc_core;
using namespace tlm;
using namespace std;
// Target
struct Target : sc_module {
    tlm_utils::simple_target_socket<Target> socket;
    SC_CTOR(Target) : socket("socket") {
        socket.register_b_transport(this, &Target::b_transport);
    }
    void b_transport(tlm_generic_payload& trans, sc_time& delay) {
        int* data = (int*) trans.get_data_ptr();
        cout << "Target received: " << *data << endl;
        delay += sc_time(10, SC_NS);
    }
};
// Initiator
struct Initiator : sc_module {
    tlm_utils::simple_initiator_socket<Initiator> socket;
    SC_CTOR(Initiator) : socket("socket") {
        SC_THREAD(thread_process);
    }
    void thread_process() {
        Read 43 lines
    }
};
```



```
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mithra@DESKTOP-6DUTPF3:~$ ls /usr/include/systemc.h
/usr/include/systemc.h
mithra@DESKTOP-6DUTPF3:~$ g++ -o test_systemc test_systemc.cpp -lsystemc
mithra@DESKTOP-6DUTPF3:~$ ./test_systemc

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Time: 0 s - Clock Triggered
Time: 1 ns - Clock Triggered
Time: 2 ns - Clock Triggered
Time: 3 ns - Clock Triggered
Time: 4 ns - Clock Triggered
mithra@DESKTOP-6DUTPF3:~$ mkdir ~/exp26
mithra@DESKTOP-6DUTPF3:~$ ^C
mithra@DESKTOP-6DUTPF3:~$ cd ~/exp26
mithra@DESKTOP-6DUTPF3:~/exp26$ pwd
/home/mithra/exp26
mithra@DESKTOP-6DUTPF3:~/exp26$ nano tlm.cpp
mithra@DESKTOP-6DUTPF3:~/exp26$ g++ tlm.cpp -lsystemc -o tlm
mithra@DESKTOP-6DUTPF3:~/exp26$ ls
tlm tlm.cpp
mithra@DESKTOP-6DUTPF3:~/exp26$ ./tlm

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Initiator sending: 555
Target received: 555
mithra@DESKTOP-6DUTPF3:~/exp26$
```